

RUSSIAN MISSION, AK, USA

WMO: 703053

Lat: 61.775N

Lon: 161.319W

Elev: 51

StdP: 14.67

Time Zone: -9.00 (NAK)

Period: 05-19

WBAN: 26654

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-36.0	-27.7	-32.5	0.9	-27.3	-28.5	1.1	-22.8	23.1	16.7	19.5	13.5	1.9	250	0.709

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.7	75.0	60.1	71.0	58.5	67.4	57.0	62.2	71.3	60.1	68.0	58.4	65.3	3.7	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
58.6	73.6	64.6	56.7	68.7	62.5	55.1	64.7	60.5	27.8	71.2	26.4	67.8	25.3	65.2	72.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.2	16.1	13.4		-37.8	80.8	8.5	4.1	-43.9	83.8	-48.9	86.2	-53.6	88.5	-59.8	91.5	
			DB	-30.6	65.3	3.4	3.5	-33.1	67.8	-35.1	69.9	-37.0	71.9	-39.5	74.4	
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	31.0	4.8	11.6	12.9	29.3	43.5	54.1	56.8	54.6	46.6	33.3	15.2	8.5
	DBStd	22.16	19.64	15.54	13.06	11.04	7.72	5.53	5.50	4.70	6.02	8.38	12.81	15.42
	HDD50	7492	1402	1076	1150	621	226	20	6	13	129	519	1044	1286
	HDD65	12409	1867	1496	1615	1071	666	328	262	323	552	984	1494	1751
	CDD50	570	0	0	0	0	25	145	217	156	27	0	0	0
	CDD65	11	0	0	0	0	0	2	8	1	0	0	0	0
	CDH74	148	0	0	0	0	4	44	94	6	0	0	0	0
	CDH80	18	0	0	0	0	0	6	12	0	0	0	0	0
Wind	WSAvg	5.2	5.8	6.0	5.5	5.2	4.8	4.8	5.0	4.5	5.0	5.6	5.3	5.4
Precipitation	PrecAvg	24.3	1.7	1.5	1.3	1.3	1.3	1.8	2.7	3.1	3.0	2.4	2.4	1.9
	PrecMax	30.2	3.7	4.4	4.9	2.7	2.9	3.1	4.7	5.6	5.6	5.1	5.4	4.3
	PrecMin	18.3	0.5	0.3	0.3	0.4	0.4	0.6	1.0	1.1	0.7	0.4	0.1	0.8
	PrecStd	3.3	0.8	1.0	0.9	0.6	0.6	0.5	0.8	0.9	1.1	1.1	1.3	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.0	39.0	42.0	57.3	73.6	79.8	82.0	74.3	65.7	52.8	41.5	40.3
		MCWB	36.9	35.8	36.2	44.2	55.7	59.5	66.0	61.0	54.8	47.0	39.1	35.1
	2%	DB	38.3	36.7	37.6	52.1	66.9	73.8	76.4	68.9	60.7	49.7	38.0	37.7
		MCWB	35.3	33.9	33.4	42.1	52.1	57.9	62.4	59.3	52.8	45.3	36.6	35.1
	5%	DB	35.4	34.7	34.8	48.0	61.8	69.8	72.1	65.8	57.5	47.1	35.0	34.2
		MCWB	33.3	32.7	32.0	39.9	49.8	56.3	60.4	57.9	51.9	44.2	33.8	32.8
	10%	DB	31.5	32.5	32.2	44.4	57.5	66.2	67.9	62.9	55.1	44.9	32.1	31.1
		MCWB	30.2	31.1	30.0	37.7	47.1	54.9	58.8	56.6	50.8	42.7	31.2	30.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.4	36.3	36.9	44.9	56.8	62.2	68.0	63.1	56.4	48.2	39.3	37.0
		MCDB	40.3	38.5	41.3	55.6	70.3	74.7	79.1	70.6	61.4	51.2	40.4	39.2
	2%	WB	35.6	34.3	34.2	42.3	53.3	59.6	63.7	60.5	54.2	46.3	36.7	34.7
		MCDB	38.2	36.0	36.8	51.3	66.0	70.7	73.8	66.1	58.5	48.6	38.2	36.9
	5%	WB	33.7	33.0	32.5	40.3	50.3	57.6	61.3	59.1	52.6	44.7	34.0	33.0
		MCDB	35.5	34.5	34.7	47.7	60.0	67.2	70.1	64.2	56.7	46.9	34.9	34.4
	10%	WB	30.6	31.3	30.0	38.1	48.1	55.6	59.2	57.5	51.2	42.7	31.2	30.3
		MCDB	32.0	32.8	32.2	43.7	56.5	64.7	66.9	62.1	54.7	44.9	32.0	31.6
Mean Daily Temperature Range	5% DB	MDBR	13.2	15.4	19.0	17.7	18.0	16.8	13.7	12.4	11.9	9.5	11.6	12.9
		MCDBR	11.9	12.0	16.3	22.4	26.1	24.4	23.3	18.1	16.2	10.3	9.2	11.3
		MCWBR	10.2	10.9	13.2	13.6	13.5	11.4	11.6	9.5	9.8	7.5	8.8	10.1
	5% WB	MCDBR	11.3	11.7	14.3	21.4	23.9	21.5	20.9	15.0	14.0	9.3	9.6	11.1
		MCWBR	9.7	10.9	11.9	13.3	12.4	10.3	10.6	8.3	9.1	7.3	9.4	10.2
Clear-Sky Solar Irradiance	taub		0.220	0.272	0.279	0.350	0.369	0.360	0.367	0.379	0.340	0.292	0.230	0.199
	taud		2.042	2.182	2.242	2.107	2.307	2.409	2.428	2.384	2.452	2.416	2.053	1.892
	Ebn at Noon		153	221	267	263	269	275	269	254	245	217	142	92
	Edh at Noon		12	21	28	40	36	33	32	30	24	17	11	8
All-Sky Solar Radiation	RadAvg		100	320	917	1453	1664	1707	1407	1097	744	356	120	47
	RadStd		15	42	97	126	156	159	127	121	80	30	20	6

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)	N/A	+3.01	+2.88	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page